

PRODUCT LITERATURE

PIRODENE CAPSULE 10MG

Each capsule contains

Piroxicam 10 mg

PIRODENE CAPSULE 20MG

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Piroxicam 20 mg

Product Description

Pirodene Capsule 10 mg: Hard gelatin capsule # 2 Ruby Red/Red

Pirodene Capsule 20 mg: Hard gelatin capsule #2 Blue/Blue

Pharmacodynamics properties

Mechanism of action/Effect:

Pharmacotherapeutic group: non-steroidal anti-inflammatory agent

Piroxicam is a non-steroidal anti-inflammatory agent with analgesic and antipyretic activity.

Piroxicam inhibits prostaglandin (thromboxane) synthesis in the platelets, rendering them less sticky. Like other NSAIDs, it acts as an uterotrophic agent by inhibiting the synthesis of prostaglandins in the uterus which are normally increased in amount in the hours before parturition.

Piroxicam also helps to promote salt and water retention by interfering with the prostaglandin-induced inhibition of both chloride re-absorption and the action of ADH.

Prostaglandins, particularly E1 and E2, are synthesised by the gastric mucosa and seem to promote integrity of that mucosa by stimulating the secretion of cytoprotective mucus. Piroxicam, by inhibiting the synthesis of these prostaglandins may lead to gastric erosions and ulceration.

Pharmacokinetics properties

Absorption:

Piroxicam is absorbed from the GI tract. Absorption is not influenced by either food or antacids.

Distribution:

Peak plasma concentrations are reached 3-5 hours after an oral dose.

Piroxicam plasma concentrations do not appear to be significantly influenced by concomitant aspirin, iron or antacids.

In man it penetrates into the synovial fluid of patients with rheumatoid arthritis, osteoarthritis and reactive synovitis, where mean concentrations are approximately 40% of those in the plasma; it is also demonstrable in synovial tissues.

Pharmacokinetics do not appear to be age related, and renal function has only a limited influence on the elimination of piroxicam, but plasma concentrations are increased in patients with severe liver dysfunction.

Biotransformation:

It is metabolised in the liver by hydroxylation and conjugation with glucuronic acid.

Elimination:

Excreted predominantly in the urine and smaller amounts in the faeces. Less than 5% of a dose is excreted unchanged. Piroxicam is extensively bound to plasma proteins (about 99%) and has a long plasma half-life of about 50 hours.

Indication:

For the management of rheumatic disorders.

Recommended Dose:

Pirodene Capsule 10mg: Adult: 1 capsule twice a day or 2 capsules once a day

Pirodene Capsule 20mg: Adult: 1 capsule a day

Route of Administration

Oral

Contraindications:

- Piroxicam should not be prescribed to patient who is more likely to develop side effects, such as those with a history of gastro-intestinal disorders associated with bleeding, or those who have had skin reactions to other medicines.
- Piroxicam should not be prescribed in association with any other NSAID or an anticoagulant.

- Hypersensitivity to the active substance or to any of the excipients, previous skin reaction (regardless of severity) to piroxicam, other NSAIDs and other medications.
- History of previous serious allergic drug reaction of any type, especially cutaneous reactions such as erythema multiforme, Stevens-Johnson syndrome, toxic epidermal necrolysis.
- Patients with active peptic ulcer, inflammatory gastrointestinal disorder or gastrointestinal bleeding.
- History of gastro-intestinal ulceration, bleeding or perforation.
- Patient history of gastrointestinal disorders that predispose to bleeding disorders such as ulcerative colitis, Crohn's disease, gastrointestinal cancers or diverticulitis.
- Concomitant use with other NSAIDs, including COX-2 selective NSAIDs and aspirin at analgesic doses.
- Concomitant use with anticoagulants.
- Piroxicam should not therefore be administered to patients in whom aspirin and other NSAIDs induce the symptoms of angioneurotic oedema, asthma, rhinitis, nasal polyps or urticaria.
- Patients with severe heart failure.

Warnings and Precautions:

- Treatment should always be initiated by a physician experienced in the treatment of rheumatic diseases.
- Use the lowest dose (no more than 20mg per day) and for the shortest duration possible. Treatment should be reviewed after 14 days.
- Always consider prescribing a gastro-protective agent.
- Undesirable effects may be minimised by using the lowest effective dose for the shortest duration necessary to control symptoms. The clinical benefit and tolerability should be reevaluated periodically and treatment should be immediately discontinued at the first appearance of cutaneous reactions or relevant gastrointestinal events.
- Cardiovascular and cerebrovascular effects
Appropriate monitoring and advice are required for patients with a history of hypertension and/or mild to moderate congestive heart failure as fluid retention and oedema have been reported in association with NSAID therapy.
- Patients with uncontrolled hypertension, congestive heart failure, established ischaemic heart disease, peripheral arterial disease, and/or cerebrovascular disease should only be treated with piroxicam after careful consideration. Similar consideration should be made before initiating longer-term treatment of patients with risk factors for cardiovascular disease (e.g. hypertension, hyperlipidemia, diabetes mellitus, and smoking).
- Cardiovascular, Renal and Hepatic Impairment
The administration of an NSAID may cause a dose dependent reduction in prostaglandin formation and precipitate renal failure. Patients at greatest risk of this reaction are those with impaired renal function, cardiac impairment, liver dysfunction, those taking diuretics and the elderly. Renal function should be monitored in these patients. In rare cases, NSAIDs may cause interstitial nephritis, glomerulonephritis, papillary necrosis and the nephrotic syndrome. Due to the renal excretion of piroxicam, patients with severely impaired renal function should be closely monitored.
- Elderly
The elderly have an increased frequency of adverse reactions to NSAIDs, especially gastrointestinal bleeding and perforation which may be fatal
- Respiratory disorders
Caution is required if administered to patients suffering from or with a previous history of bronchial asthma since NSAIDs have been reported to precipitate bronchospasm in such patients. Due to reports of adverse eye findings with NSAIDs, it is recommended that patients who develop visual complaints during treatment with piroxicam have ophthalmic evaluation.
- Gastrointestinal bleeding, ulceration and perforation
NSAIDs, including piroxicam, can cause serious gastrointestinal events including bleeding, ulceration, and perforation of the stomach, small intestine or large intestine, which can be fatal. These serious adverse events can occur at any time, with or without warning symptoms, in patients treated with NSAIDs.

NSAID exposures of both short and long duration have an increased risk of serious GI event. Evidence from observational studies suggests that piroxicam may be associated with a high risk of serious gastrointestinal toxicity, relative to other NSAIDs.

Patients with significant risk factors for serious GI events should be treated with piroxicam only after careful consideration.

The possible need for combination therapy with gastro-protective agents (e.g. misoprostol or proton pump inhibitors) should be carefully considered and also for patients requiring concomitant low dose aspirin, or other drugs

likely to increase gastrointestinal risk.

- **Serious GI Complications**

Identification of at-risk subjects: The risk for developing serious GI complications increases with age. Age over 70 years is associated with high risk of complications. The administration to patients older than 80 years should be avoided.

The risk of GI bleeding, ulceration or perforation is higher with increasing NSAID doses, in patients with a history of ulcer, particularly if complicated with haemorrhage or perforation and in the elderly. These patients should commence treatment on the lowest dose available.

When GI bleeding or ulceration occurs in patients receiving piroxicam the treatment should be withdrawn.

NSAIDs should be given with care to patients with a history of gastrointestinal disease (ulcerative colitis, Crohn's disease) as these conditions may be exacerbated.

Patients taking concomitant oral corticosteroids, selective serotonin reuptake inhibitors (SSRIs) or anti-platelet agents such as low-dose aspirin are at increased risk of serious GI complications. As with other NSAIDs, the use of piroxicam in combination with protective agents (e.g. misoprostil and proton pump inhibitors) must be considered for these at-risk patients.

Patients and physicians should remain alerted for signs and symptoms of GI ulceration and/or bleeding during piroxicam treatment. Patients should be asked to report any new or unusual abdominal symptom during treatment. If a gastrointestinal complication is suspected during treatment, piroxicam should be discontinued immediately and additional clinical evaluation and treatment should be considered.

- **SLE and mixed connective tissue disease**

In patients with systemic lupus erythematosus (SLE) and mixed connective tissue disorders there may be an increased risk of aseptic meningitis

- **Dermatological**

Serious skin reactions, some of them fatal, including exfoliative dermatitis, Stevens-Johnson syndrome, and toxic epidermal necrolysis, have been reported very rarely in association with the use of NSAIDs.

Evidence from observational studies suggests that piroxicam may be associated with a higher risk of serious skin reactions than other NSAIDs. Life-threatening cutaneous reactions Stevens-Johnson syndrome (SJS) and toxic epidermal necrolysis (TEN) have been reported with the use of piroxicam. Patients should be advised of the signs and symptoms and monitored closely for skin reactions. The highest risk for occurrence of SJS or TEN is within the first weeks of treatment. If symptoms or signs of SJS or TEN (e.g. progressive skin rash often with blisters or mucosal lesions) are present, piroxicam treatment should be discontinued. The best results in managing SJS and TEN come from early diagnosis and immediate discontinuation of any suspect drug. Early withdrawal is associated with a better prognosis. If the patient has developed SJS or TEN with the use of piroxicam, piroxicam must not be re-started in this patient at any time.

Patients appear to be at a highest risk of these reactions early in the course of therapy, the onset of the reaction occurring in the majority of cases within the first month of treatment. Piroxicam should be discontinued at the first appearance of skin rash, mucosal lesions, or any other sign of hypersensitivity.

- **Impaired female fertility**

The use of piroxicam may impair female fertility and is not recommended in women attempting to conceive. In women who have difficulties conceiving or who are undergoing investigation of infertility, withdrawal of Piroxicam should be considered.

- **Headache**

Where analgesics are used long-term (>3 months) with administration every two days or more frequently, headache may develop or worsen. Headache induced by overuse of analgesics (MOH medication-overuse headache) should not be treated by dose increase. In such cases, the use of analgesics should be discontinued in consultation with the doctor.

- **Lactose**

Piroxicam contain lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicinal product.

Interactions with Other Medicament:

Antacids: Concomitant administration of antacids had no effect on piroxicam plasma levels.

Anti-coagulants: NSAIDs, including piroxicam, may enhance the effects of anticoagulants, such as warfarin. Therefore, the use of piroxicam with concomitant anticoagulant such as warfarin should be avoided

Anti-platelet agents and selective serotonin reuptake inhibitors (SSRIs):

Increased risk of gastrointestinal bleeding

Aspirin and other Non-Steroidal Anti-Inflammatory Drugs: Piroxicam, like other non-steroidal anti-inflammatory drugs decreases platelet aggregation and prolongs bleeding time. This effect should be kept in mind when bleeding times are determined.

As with other NSAIDs, the use of piroxicam together with acetylsalicylic acid or concomitant use with other NSAIDs, including other piroxicam formulations, must be avoided, since data are inadequate to show that combinations produce greater improvement than that achieved with piroxicam alone; moreover, the potential for adverse reactions is enhanced. Human studies have shown that concomitant use of piroxicam and acetylsalicylic acid reduces the plasma piroxicam concentration to about 80% of the usual value.

Cardiac glycosides: NSAIDs may exacerbate cardiac failure, reduce GFR and increase plasma glycoside levels.

Ciclosporin, Tacrolimus: possible increased risk of nephrotoxicity when NSAIDs are given with ciclosporin or tacrolimus.

Cimetidine: Results of two separate studies indicate a slight but significant increase in absorption of piroxicam following cimetidine administration but no significant changes in elimination rate constants or half-life. The small increase in absorption is unlikely to be clinically significant.

Corticosteroids: increased risk of gastrointestinal ulceration or bleeding

Digoxin, Digitoxin: Concurrent therapy with piroxicam and digoxin, or piroxicam and digitoxin, did not affect the plasma levels of either drug.

Anti-hypertensives including diuretics, angiotensin-converting enzyme (ACE) inhibitors, angiotensin II antagonists (AIIA) and beta-blockers:

NSAIDs can reduce the efficacy of diuretics and other anti-hypertensive drugs including ACE inhibitors, AIIA and beta-blockers. In patients with impaired renal function (e.g. dehydrated patients or elderly patients with the renal function compromised), the co-administration of an ACE inhibitor or an AIIA and/or diuretics with a cyclo-oxygenase inhibitor can increase the deterioration of the renal function, including the possibility of acute renal failure, which is usually reversible.

The occurrence of these interactions should be considered in patients taking piroxicam with an ACE inhibitor or an AIIA and/or diuretics. Therefore, the concomitant administration of these drugs should be done with caution, especially in elderly patients. Patients should be adequately hydrated and the need to monitor the renal function should be assessed in the beginning of the concomitant treatment and periodically thereafter.

Highly protein-bound drugs: Piroxicam is highly protein-bound and therefore might be expected to displace other protein-bound drugs. The physician should closely monitor patients for change when administering Piroxicam to patients on highly protein-bound drugs.

Lithium: Non-steroidal anti-inflammatory drugs, including Piroxicam, have been reported to increase steady state plasma lithium levels. It is recommended that these levels are monitored when initiating, adjusting and discontinuing Piroxicam.

Piroxicam, like other non-steroidal anti-inflammatory drugs, may interact with the following drugs / classes of therapeutic agents:

Antihypertensives -antagonism of the hypotensive effect

Quinolone antibiotics - possible increased risk of convulsions

Mifepristone - NSAIDs could interfere with mifepristone-mediated termination of pregnancy.

Methotrexate - Reduced excretion of methotrexate, possibly leading to acute toxicity. When methotrexate is administered concurrently with NSAIDs, including piroxicam, NSAIDs may decrease elimination of methotrexate resulting in increased plasma levels of methotrexate. Caution is advised, especially in patients receiving high doses of methotrexate.

Pregnancy/Reproduction

Pregnancy:

Congenital abnormalities have been reported in association with NSAID administration in man; however, these are low in frequency and do not appear to follow any discernible pattern. Piroxicam inhibits prostaglandin synthesis and release through a reversible inhibition of the cyclo-oxygenase enzyme.

This effect, as with other NSAIDs has been associated with an increased incidence of dystocia and delayed parturition in pregnant animals when drug administration was continued into late pregnancy. In view of the known effects of NSAIDs on the foetal cardiovascular system (risk of closure of the ductus arteriosus), use in the last trimester of pregnancy is contraindicated. The onset of labour may be delayed and the duration increased with an increased bleeding tendency in both mother and child.

NSAIDs should not be used during the first two trimesters of pregnancy or labour unless the potential benefit to the patient outweighs the potential risk to the foetus.

Breast-feeding:

In limited studies so far available, NSAIDs can appear in breast milk in very low concentrations. A study indicated that piroxicam appears in breast milk at about 1-3% of the maternal plasma concentrations. No accumulation of piroxicam occurred in milk relative to that in plasma during treatment for up to 52 days. NSAIDs should, if possible, be avoided when breastfeeding.

Fertility

See section Warning & Precautions for use regarding fertility.

Side Effects

Results from clinical trials involving approximately 2300 patients (of whom about 400 were treated for more than one year and 170 for more than two years) indicate that about 30% of patients reported side effects at a dose of 20 mg/day. This increased with doses of 30 - 40 mg/day.

More Common Reactions (more than 3%)

Gastrointestinal

These have been the most frequent side effects, occurring in about 20% of patients. Approximately 5% discontinued therapy with an overall incidence of peptic ulcer of about 1%.

The gastrointestinal side effects included abdominal discomfort (5.7%), flatulence (5.2%), nausea (4.8%), abdominal pain (4.7%), epigastric distress (4.1%), constipation (3.8%) and diarrhoea (3.2%).

Central Nervous System

Dizziness (4.1%), headache (4.1%).

Less Common Reactions (less than 3%)

Auditory and Vestibular

Tinnitus, vertigo, deafness.

Laboratory Abnormalities

Elevated levels of hepatic enzymes (LDH, alkaline phosphatase, transaminases); elevation of blood urea nitrogen (BUN) and serum creatinine; depression of levels of haemoglobin and haematocrit; depression of levels of serum proteins, platelet and white blood cell count.

Cardiovascular

Hypertension, tachycardia, palpitations.

Skin and Subcutaneous Tissue Disorders

Skin rash (2.4%), pruritus (1.1%), onycholysis, alopecia. Photoallergic reactions have been infrequently associated with therapy. As with other NSAIDs, drug reaction with eosinophilia and systemic symptoms (DRESS syndrome), drugs, toxic epidermal necrolysis (Lyell's disease) and Stevens-Johnson syndrome may develop in rare cases. Vesicular bullous reactions have been reported rarely.

Gastrointestinal

Anorexia, vomiting, indigestion, pancreatitis, hepatitis.

Central Nervous System

Sedation, drowsiness (2.1%), others (each less than 1%) include amnesia, anxiety, depression, malaise, hallucinations, insomnia, dream abnormalities, nervousness, paraesthesia, personality change, tremors, akathisia.

Genitourinary

Oedema (2.7%), others (less than 1%) includes dysuria, urinary frequency, haematuria, oliguria, menorrhagia.

Eyes, Nose, and Throat

Stomatitis (1%), blurred vision, eye irritation/swelling, epistaxis, glossitis.

Haematological

Decreases in haemoglobin and haematocrit, unassociated with obvious gastrointestinal bleeding, have occurred. Anaemia has been reported. Thrombocytopenic and non-thrombocytopenic purpura (Henoch-Schönlein), petechial rash, ecchymosis, leucopenia and eosinophilia have been reported. Rare cases of aplastic anaemia and haemolytic anaemia are also reported.

Miscellaneous (each less than 1%)

Breathlessness, chest pain, hyperglycaemia, hypoglycaemia, thirst, chills, sweating, flushing, increased appetite, weight increase or decrease. Rare anecdotal reports of positive antinuclear antibodies.

Symptoms and Treatment of Overdose

a) Symptoms

Symptoms include headache, nausea, vomiting, epigastric pain, gastrointestinal bleeding, rarely diarrhoea, disorientation, excitation, coma, drowsiness, dizziness, tinnitus, fainting, occasionally and convulsions. In cases of significant poisoning acute renal failure and liver damage are possible.

b) Therapeutic measure

Patients should be treated symptomatically as required.

Within one hour of ingestion of a potentially toxic amount, activated charcoal should be considered. Alternatively, in adults, gastric lavage should be considered within one hour of ingestion of a potentially life-threatening overdose.

Good urine output should be ensured.

Renal and liver function should be closely monitored.

Patients should be observed for at least four hours after ingestion of potentially toxic amounts.

Frequent or prolonged convulsions should be treated with intravenous diazepam.

Other measures may be indicated by the patient's clinical condition.

Packing

Piordene Capsule 10 mg:

Packing in blister, 10 capsules per blister.

Pack size: 10's, 30's, 100's and 1000's capsules.

Piordene Capsule 20 mg:

Packing in blister, 10 capsules per blister.

Pack size: 10's, 30's, 100's and 1000's capsules.

Storage conditions

Store below 30°C in a dry place, protected from light.

Keep away from reach of children.

Expiry Date

Piordene Capsule 10 mg: 3 years from date of manufacture

Piordene Capsule 20 mg: 3 years from date of manufacture

Name and Address of Product Registration Holder/Distributor:

ZONTRON PHARMACEUTICALS SDN. BHD. (445695-T)

Lot 10 & 11, PERDA Industrial Park,
Lorong IKS Simpang Ampat B,
14100 Simpang Ampat, S.P.S.,
Pulau Pinang, Malaysia

Name and Address of Manufacturer:

TERAPUTICS SDN. BHD. (590500-W)

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Malaysian Drug Registration No. :

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